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Case 5-2022: A 65-Year-Old Woman with Rapidly Progressive Weakness in the Right Arm and Recurrent Falls

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PRESENTATION OF CASE

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Dr. Marina Avetisyan (Neurology, Brigham and Women's Hospital): A 65-year-old woman was admitted to this hospital because of rapidly progressive weakness in the right arm and recurrent falls.

Twelve years before this admission, the patient noted worsening gait and balance disturbance, which slowly progressed, and falls began to occur 5 years before this admission. Her medical history included measles encephalitis, which had been diagnosed when she was 6 years of age and had resulted in weakness on the left side, severe vision and hearing impairment, and gait and balance disturbance. After rehabilitation therapy, her balance and gait apparently normalized, but she had mild residual weakness on the left side, which resulted in a change from left-hand to right-hand dominance.

Two years before this admission, the patient was evaluated in the neurology clinic of this hospital. She reported difficulties with balance, which worsened with head turning. She reported no disequilibrium, vertigo, back pain, paresthesia, or numbness. On examination, her cognitive function was intact, and although her speech was dysarthric and hypophonic, these abnormalities could have been associated with her hearing loss. The cranial-nerve examination revealed abnormal eye movements, with square-wave jerks, saccadic pursuit, and bilateral horizontal nystagmus. The hearing and vision were severely reduced. The muscle bulk was decreased throughout, and tone was normal. Motor strength was full and symmetric in the arms and legs bilaterally, and contractures were present in the fingers of the left hand. Reflexes were reduced or absent throughout, and plantar reflexes were extensor bilaterally. Vibratory sensation was reduced at the great toes but was otherwise normal. The patient had appendicular dysmetria on both finger-to-nose testing and heel-to-shin testing that was greater on the left side than on the right side. Her gait was wide-based, and she was unable to walk with a tandem gait. The Romberg sign was present.

Needle electromyographic findings and the results of nerve-conduction studies were normal, as were findings on magnetic resonance imaging (MRI) of the head, performed before and after the administration of intravenous contrast material. MRI of the cervical and thoracic spine revealed mild multilevel degenerative disk disease without clinically significant spinal-canal stenosis. A lumbar puncture was performed; analysis of the cerebrospinal fluid (CSF) showed 0 nucleated cells per microliter (reference range, 0 to 5), and oligoclonal bands were absent. Tests of the CSF and serum for antibodies associated with autoimmune and paraneoplastic encephalitis were negative (Table 1). The patient underwent physical therapy, and her symptoms stabilized.

Three months before this admission, weakness in the right hand and right arm developed. The patient initially attributed the weakness to fatigue, since it had occurred during a time of increased physical activity associated with moving to a different apartment. However, the weakness did not abate, and 2 months before this admission, she was again evaluated in the neurology clinic of this hospital. She noted difficulty with communicating in sign language and with holding objects in her right hand because of weakness and had started using her left hand to eat. She reported no neck, arm, or leg pain and no sensory loss. Her gait and balance were stable, with no recent falls.

On examination, there was reduced facial expression and reduced muscle bulk and tone throughout. The right shoulder was dropped, with weakness in the right trapezius and reduced active shoulder abduction that was limited to 70 degrees, without pain on shoulder abduction. There was predominantly proximal weakness in the right arm; the muscle strength was 4–/5 on shoulder abduction and external rotation, 4+/5 on elbow flexion, and 4/5 on elbow extension, wrist extension, and abductor digiti minimi and finger extension. The strength in the right leg was normal. There was subtle weakness on the left side that involved wrist and finger extension and great toe extension. Deep-tendon reflexes were 2+ at the right biceps and 1+ at the left biceps and were otherwise areflexic throughout. Sensation was intact in the arms, but there was decreased vibratory sensation and proprioception at the great toes; sensation was

otherwise normal. Moderate appendicular dysmetria was present in the arms and legs. The patient's gait, which was assessed while she was using her white cane (the type of cane commonly used by blind or visually impaired persons), was wide-based; she took excessively large steps and reached out once to touch the wall. Neurologic examination was otherwise unchanged from the examination performed 2 years earlier. Additional studies were obtained.

Dr. Pamela W. Schaefer: MRI of the head, performed without the administration of intravenous contrast material, was reported to be normal. MRI of the cervical spine was also performed without the administration of contrast material; T2-weighted fluid-attenuated inversion recovery (FLAIR) imaging showed signal hyperintensity in the suboccipital and paraspinal soft tissues that was consistent with edema. Areas of intrinsic hyperintense signal that were consistent with fatty atrophy were seen on T1-weighted imaging. There were diffuse degenerative changes with mild spinal-canal stenosis at C4–C5, without cord compression.

Dr. Avetisyan: Repeat needle electromyography revealed no recorded motor units in the right deltoid or infraspinatus muscles, with complex repetitive discharges in the right cervical paraspinal and deltoid muscles. Nerve-conduction studies revealed mild reduction in the amplitudes of the bilateral median and right ulnar sensory responses; the sensory responses of the right radial, bilateral antebrachial cutaneous, and bilateral medial antebrachial cutaneous nerves were normal, as were the motor responses of the right median and right ulnar nerves.

During the ensuing 2 months, the weakness in the right arm progressed rapidly, which caused the patient to have difficulty ambulating with her white cane (she was unable to walk with a standard cane), making walking unsafe after frequent falls and rendering her practically bedbound. She had no pain or numbness in her arm. She was ultimately unable to care for herself at home and was admitted to this hospital for further evaluation.

Other medical history included hypercholesterolemia and hypothyroidism. Medications included ascorbic acid, vitamin B complex, coenzyme Q10, vitamin E, calcium carbonate and cholecalciferol, and levothyroxine. There were no

Table 1. Laboratory Data.*

Variable	Reference Range, Adults†	2 Yr before Admission	On Admission
Blood			
Hemoglobin (g/dl)	12.0–16.0	—	12.9
Hematocrit (%)	41.0–53.0	—	39.2
White-cell count (per μ l)	4500–11,000	—	5670
Differential count (per μ l)			
Neutrophils	1800–7700	—	3710
Lymphocytes	1000–4800	—	1430
Monocytes	200–1200	—	450
Eosinophils	0–900	—	40
Basophils	0–300	—	30
Granulocytes	0–100	—	10
Platelet count (per μ l)	150,000–450,000	—	247,000
Glucose (mg/dl)	70–110	—	88
Creatinine (mg/dl)	0.60–1.50	—	0.52
Urea nitrogen (mg/dl)	8–25	—	14
Aspartate aminotransferase (U/liter)	9–32	—	38
Alanine aminotransferase (U/liter)	7–33	—	33
Alkaline phosphatase (U/liter)	30–100	—	56
Total bilirubin (mg/dl)	0.0–1.0	—	0.5
Albumin (g/dl)	3.3–5.0	—	3.4
Globulin (g/dl)	1.9–4.1	—	2.5
Thyrotropin (μ U/ml)	0.40–5.00	—	9.98
Free thyroxine (ng/dl)	0.9–1.9	—	1.2
Creatine kinase (U/liter)	40–150	—	132
Vitamin B ₁₂ (pg/ml)	251–911	—	1223
Folate (ng/ml)	>3.9	—	8.1
Antineutrophil cytoplasmic antibody	Negative	Negative	—
Antinuclear antibody	Negative	—	Negative
IgG (mg/dl)	614–1295	875	—
IgA (mg/dl)	69–309	178	—
IgM (mg/dl)	53–334	126	—
Serum protein electrophoresis	—	Normal pattern	—
<i>Treponema pallidum</i> antibody	Negative	Negative	—
Acetylcholine receptor binding antibody (mmol/liter)	≤ 0.02	0	—
Acetylcholine receptor ganglionic neuronal antibody (mmol/liter)	≤ 0.02	0	—
N-methyl-D-aspartate receptor antibody	Negative	Negative	—
Neuronal voltage-gated potassium channel–complex antibody (nmol/liter)	≤ 0.02	0	—
Leucine-rich, glioma-inactivated 1 IgG	Negative	Negative	—
Contactin-associated protein–like 2 IgG	Negative	Negative	—

Table 1. (Continued.)

Variable	Reference Range, Adults†	2 Yr before Admission	On Admission
Glutamic acid decarboxylase 65 antibody (nmol/liter)	≤0.02	0	—
γ-Aminobutyric acid type B receptor antibody	Negative	Negative	—
α-Amino-3-hydroxy-5-methyl-4-isoxazolepropionic acid receptor antibody	Negative	Negative	—
Antineuronal nuclear antibody type 1, type 2, or type 3	<1:240	Negative	—
Antiglial nuclear antibody type 1	<1:240	Negative	—
Amphiphysin antibody	<1:240	Negative	—
Purkinje cytoplasmic antibody type 1 or type 2	<1:240	Negative	—
Purkinje cytoplasmic–Tr antibody	<1:240	Negative	—
Calcium-channel binding antibody	≤0.03	0	—
Anti–collapsin response mediator protein 5 IgG	<1:240	Negative	—
Cerebrospinal fluid			
Red-cell count (per μl)	0–5	0	—
Total nucleated cells (per μl)	0–5	0	—
Differential count (%)			
Lymphocytes	0–100	0	—
Reactive lymphocytes	0	0	—
Plasma cells	0	0	—
Monocytes	0–100	0	—
Neutrophils	0–100	0	—
Protein (mg/dl)	5–55	15	—
Glucose (mg/dl)	50–75	63	—
Oligoclonal bands	—	No banding seen	—
N-methyl-D-aspartate receptor antibody	Negative	Negative	—
Neuronal voltage-gated potassium channel–complex antibody (nmol/liter)	0.00–0.02	0.00	—
Leucine-rich, glioma-inactivated 1 IgG	Negative	Negative	—
Contactin-associated protein–like 2 IgG	Negative	Negative	—
Glutamic acid decarboxylase 65 antibody (nmol/liter)	≤0.02	0.00	—
γ-Aminobutyric acid type B receptor antibody	Negative	Negative	—
Antineuronal nuclear antibody type 1, type 2, or type 3	<1:2	Negative	—
Antiglial nuclear antibody type 1	<1:2	Negative	—
Purkinje cytoplasmic antibody type 1	<1:2	Negative	—
Purkinje cytoplasmic–Tr antibody	<1:2	Negative	—
Amphiphysin antibody	<1:2	Negative	—
Anti–collapsin response mediator protein 5 IgG	<1:2	Negative	—

* To convert the values for glucose to millimoles per liter, multiply by 0.05551. To convert the values for creatinine to micromoles per liter, multiply by 88.4. To convert the values for urea nitrogen to millimoles per liter, multiply by 0.357. To convert the values for bilirubin to micromoles per liter, multiply by 17.1. To convert the values for free thyroxine to picomoles per liter, multiply by 12.87. To convert the values for vitamin B₁₂ to picomoles per liter, multiply by 0.7378. To convert the values for folate to nanomoles per liter, multiply by 2.266.

† Reference values are affected by many variables, including the patient population and the laboratory methods used. The ranges used at Massachusetts General Hospital are for adults who are not pregnant and do not have medical conditions that could affect the results. They may therefore not be appropriate for all patients.

known drug allergies. The patient lived alone and worked for a nonprofit agency. She was a lifelong nonsmoker, drank four glasses of wine per week, and did not use illicit drugs. She reported that her father had died from Creutzfeldt-Jakob disease when he was 71 years of age, 7 months after the onset of symptoms. Her mother had Alzheimer's disease, which was diagnosed when she was in her 60s; she was currently living in a nursing home. The patient had four siblings; one had died at 46 years of age from an intracerebral hemorrhage, and her remaining three siblings were well.

On examination, the temperature was 35.7°C, the blood pressure 116/64 mm Hg, the pulse 61 beats per minute, the respiratory rate 18 breaths per minute, and the oxygen saturation 100% while the patient was breathing ambient air. The body-mass index (the weight in kilograms divided by the square of the height in meters) was 15.8. The patient appeared emaciated but was able to sit up in a chair and hold her head up. Strength in her right arm had decreased to 2/5 on shoulder shrug, shoulder abduction, elbow flexion, and elbow extension, with 3/5 strength on wrist extension and finger flexion and extension. There was new proximal weakness in the right leg, with 4/5 strength on right hip flexion. The strength in the left arm was unchanged. Deep-tendon reflexes were 3+ at the right biceps and 2+ at the brachioradialis and were otherwise absent throughout. She had ataxia as well as occasional jerks that involved the legs (with the left leg affected to a greater extent than the right leg) and were possibly stimulus-induced. The remainder of the neurologic examination was unchanged from the last visit.

Dr. Schaefer: MRI of the head, performed before and after the administration of intravenous contrast material, revealed a new linear signal on susceptibility-weighted imaging but also a marked decrease in subtle restricted diffusion on diffusion-weighted imaging and hyperintensity on FLAIR imaging (identified in retrospect on the previous MRI) in the left frontal and left parietal lobes (Fig. 1). MRI of the whole spine, performed before and after the administration of intravenous contrast material, was unchanged from previous imaging.

Dr. Avetisyan: The creatine kinase level was normal. Other laboratory test results are shown in Table 1. A diagnostic test was performed.

DIFFERENTIAL DIAGNOSIS

Dr. Christopher D. Stephen: I was involved in the care of this patient, and I am aware of the diagnosis in this case. I was the first neurologist to see this patient after a long interval; on examination, it appeared that her longstanding ataxia was generally similar to that seen at her assessment 2 years earlier, albeit having perhaps slightly worsened symptomatically over time, and should be considered separately. However, the more pressing matter is the development of subacute and subsequently rapidly progressive weakness in her right arm, which later spread to involve her right leg. The time course in which this weakness developed is important, since it provides insight into the types of processes that can lead to the rapid progression of symptoms and signs seen in this patient.¹⁻³

RAPIDLY PROGRESSIVE WEAKNESS

First, in this patient with rapidly progressive weakness in the right arm and right leg, we must consider whether the weakness is caused by a central process or a peripheral process (Fig. 2). The pyramidal distribution of weakness in the arm, followed by weakness in the leg and findings on electromyography, are suggestive of a central process. Despite the presence of global muscular atrophy, the paretic limbs are not affected to a greater extent than the nonparetic limbs, and there are no signs that the patient has a primary muscle disorder, given the upper motor neuron features, the normal creatine kinase level, and the absence of associated myopathic findings on electromyography.

The absence of fatigable weakness, of symptoms that are more global or symmetric, and of ocular and bulbar weakness, as well as the presence of increased reflexes, make a neuromuscular-junction disorder highly unlikely in this case. Although progressive peripheral nerve and root disorders and other lower motor neuron conditions are common causes of rapidly progressive weakness, the features in this patient that could be interpreted as lower motor neuron signs are longstanding, and no neuropathic features are present on electromyography, with features instead suggestive of a central process.⁵

Anterior-horn-cell disease is important to consider in this case, since amyotrophic lateral sclerosis and primary lateral sclerosis can be

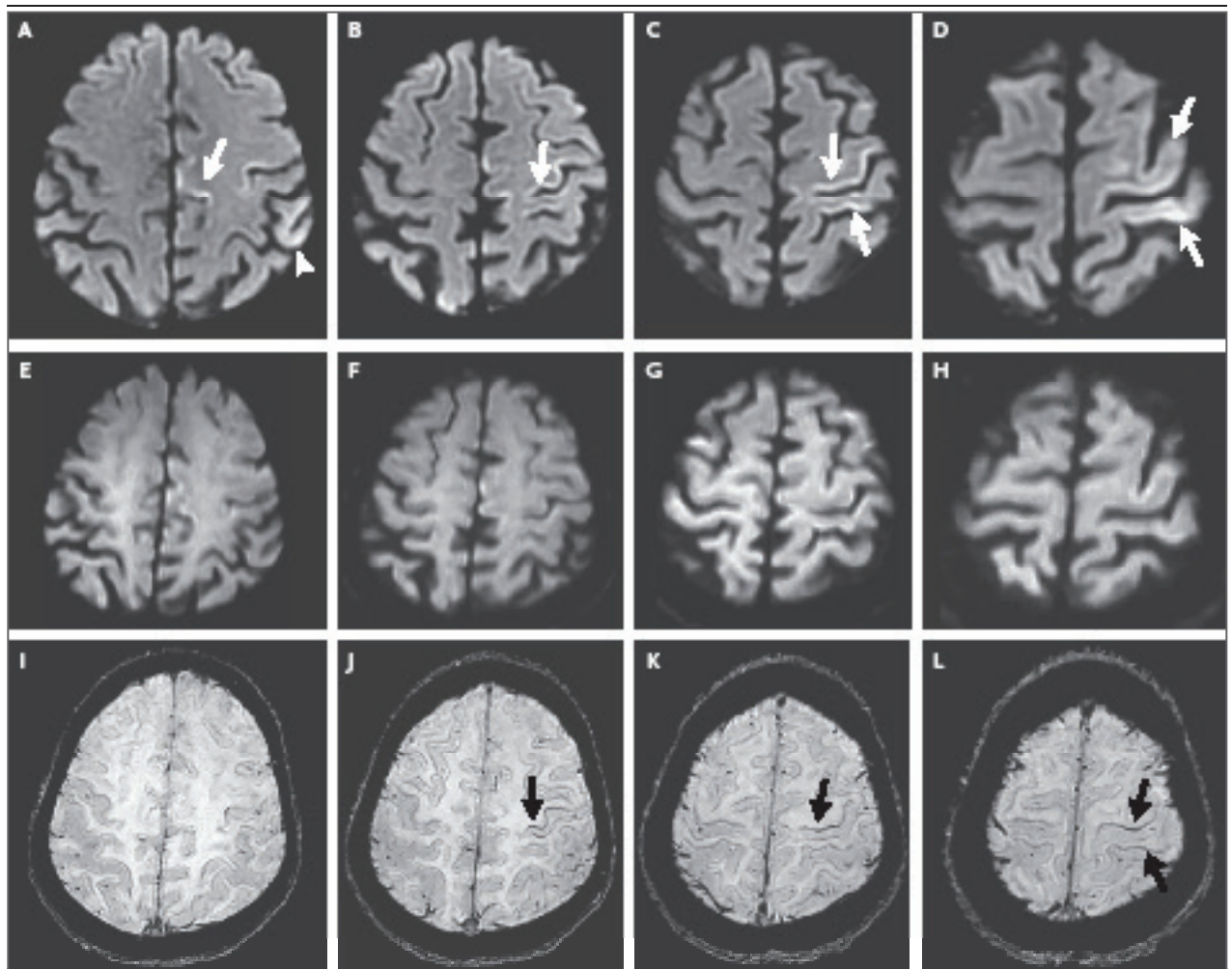


Figure 1. MRI of the Head.

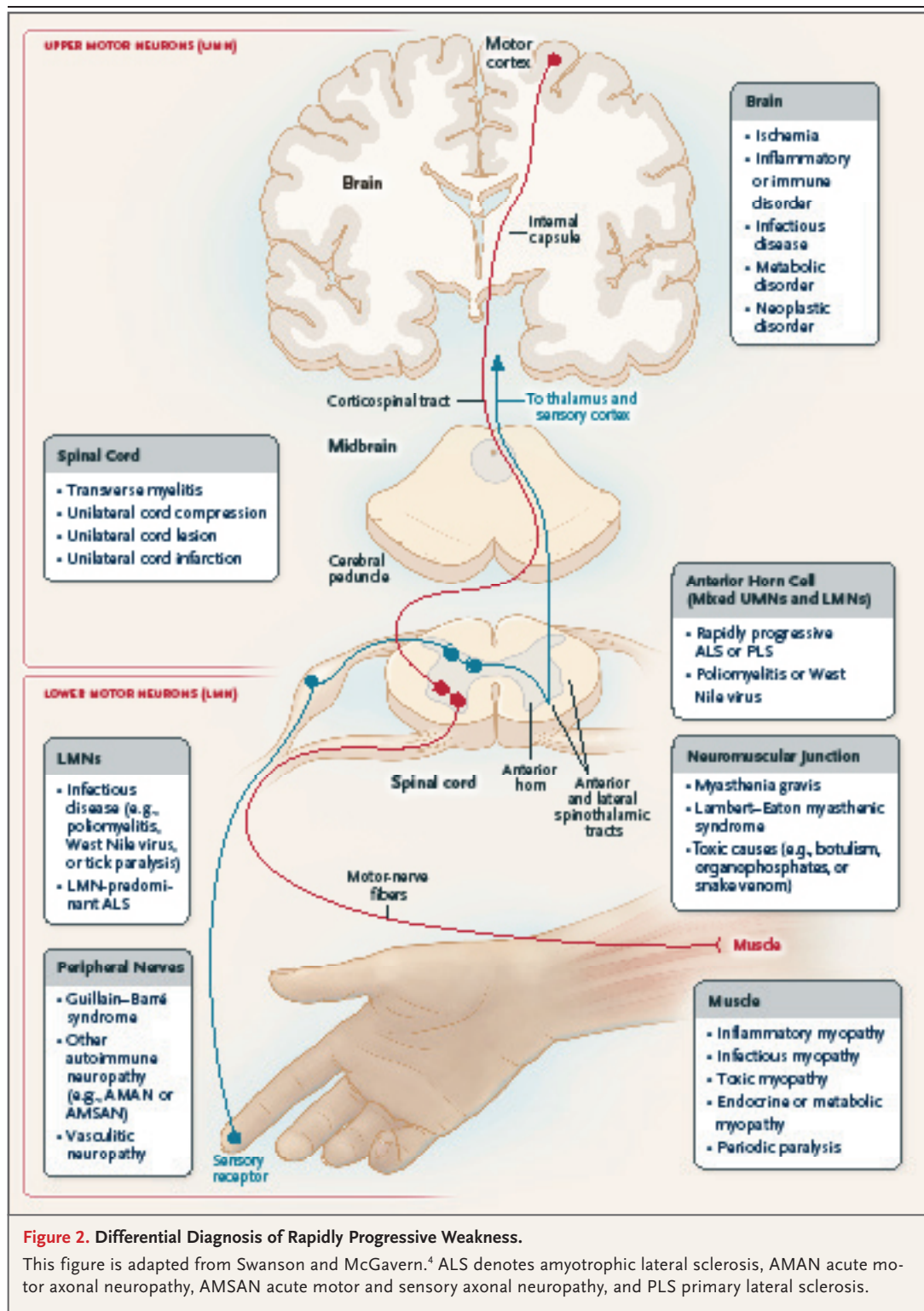
Diffusion-weighted images obtained 20 days before this admission show high signal intensity, consistent with restricted diffusion, in the left precentral and postcentral gyri (Panels B through D, arrows) as well as in the left superior frontal gyrus (Panel A, arrow) and the left supramarginal gyrus (Panel A, arrowhead). Diffusion-weighted images obtained 2 days after this admission show a marked decrease in the visibility of the foci of restricted diffusion that were previously seen in the cortex (Panels E through H). Susceptibility-weighted images obtained 2 days after admission show linear cortical susceptibility in the left precentral and postcentral gyri where restricted diffusion was previously seen (Panels I through L, arrows).

rapidly progressive and often start in a limb, with subsequent spread.⁶ However, the pyramidal distribution of weakness and the absence of acute or chronic neurogenic features on electromyography (e.g., fibrillations, sharp waves, spontaneous activity, or change in motor-unit action potential) make anterior-horn-cell disease unlikely. Diseases at the level of the spinal cord — which are generally related to compressive, inflammatory, or ischemic insults to the right side of the spinal cord (in the case of unilateral weakness) — can be ruled out on the basis

of the negative findings on spine imaging in this case.

DISEASES OF THE BRAIN

This leads us to a localization of this patient's disease at the level of the brain, given the imaging findings involving the left frontal and parietal lobes. Such imaging findings are most often associated with ischemia, whether related to stroke or to cerebral venous thrombosis; however, the subacute, and later acute, tempo of progression that occurred in this patient makes



ischemia unlikely. There is no indication of seizures or hypoxia, either of which can result in cortical-based changes.

Other important considerations in this case are infectious or inflammatory diseases that can cause cerebritis, such as herpes simplex encephalitis.

litis, rabies, and autoimmune encephalitis, although these diseases are almost always accompanied by cognitive decline or encephalopathy.^{7,8} This group of disorders also includes prion disease, which must be considered, particularly given the patient's family history. However, this patient's presentation, with weakness as the primary feature, is highly atypical for prion disease. The absence of features that would suggest an active infection and the absence of an elevated protein level in the CSF, which could indicate an autoimmune disorder, make an autoimmune or infectious encephalitis-like process unlikely, albeit not impossible. There are no metabolic derangements, and other inflammatory conditions, such as MELAS (mitochondrial encephalomyopathy, lactic acidosis, and stroke-like episodes), would be unlikely.

MYOCLONUS

In addition to weakness, this patient has new jerking movements in the legs that are suggestive of myoclonus. When evaluating a patient with suspected myoclonus, it is important to first determine whether the process is cortical or subcortical or whether it has a spinal or more peripheral origin (Fig. 3).^{9,10} Certain findings in this patient do not suggest segmental or propriospinal myoclonus; these include the absence of a spinal lesion, the fact that the distribution of jerking involves the legs and not the hips or trunk (whereas propriospinal myoclonus typically involves axial flexion jerks), and the patient's electromyographic features.^{11,12} In addition, the location of the jerking movements is not within a plexus, root, or individual peripheral nerve distribution, locations that would suggest peripheral myoclonus.¹³

Subcortical myoclonus originates between the cortex and the spinal cord, is typically bilateral, and may be global in distribution. This patient has no features that would suggest a brain-stem origin; in addition, she is not receiving any medications that could be associated with subcortical myoclonus and has no reported toxic exposures or metabolic features that could result in such a diagnosis. The findings seen on this patient's imaging, which is based primarily in the cortical region, are instead suggestive of cortical myoclonus.

Cortical myoclonus is caused by intrinsic cortical hyperexcitability, which may be focal, segmental, or multifocal.¹³ Although cortical myo-

clonus tends to involve the hands and face most often, it is a likely diagnosis in this case, given that this patient's muscle jerks occur primarily in the paretic limbs and appear to be stimulus-sensitive and that the imaging findings are localized to the cortex. The patient does not have associated epilepsy or encephalopathy, and the clinical progression is more rapid than that typically associated with any of the myoclonic epilepsy syndromes, which are not manifested by similar weakness.¹⁴ No metabolic cause has been identified in this patient; a metabolic cause would usually produce a generalized myoclonus and would not be associated with unilateral weakness. Infectious or paraneoplastic encephalopathies are generally not associated with unilateral weakness and would be unlikely diagnoses in this case.^{8,15} This patient's progression is much faster than would be seen in a degenerative basal ganglia disorder such as corticobasal syndrome, which can be manifested by strikingly unilateral dystonia and is frequently associated with myoclonus and cortical sensory loss.

CREUTZFELDT-JAKOB DISEASE

We must then consider the unusual nature of this patient's presentation and the associated rapid progression in the context of her family history of Creutzfeldt-Jakob disease — a rapidly progressive disorder that is typically associated with the development of dementia, ataxia, and myoclonus, inexorably progressing to akinetic mutism and death. We do not know what symptoms this patient's father had, and because Creutzfeldt-Jakob disease is typically a disorder of unknown cause, it is important to be aware that genetic forms of Creutzfeldt-Jakob disease exist and that they manifest in an autosomal dominant manner, which would be compatible with inheritance from her father.¹⁶ A diagnosis of Creutzfeldt-Jakob disease would also be compatible with the cortical ribboning (a ribbonlike signal hyperintensity involving the cerebral cortical gyri) seen on this patient's imaging, which is an ominous sign and is highly suggestive of prion disease.^{17,18} Although rapidly progressive weakness is an atypical finding in patients with Creutzfeldt-Jakob disease, other prion disorders such as the Gerstmann-Sträussler-Scheinker syndrome can be associated with a neuromuscular phenotype involving weakness, areflexia, and paresthesia, albeit with far slower progression

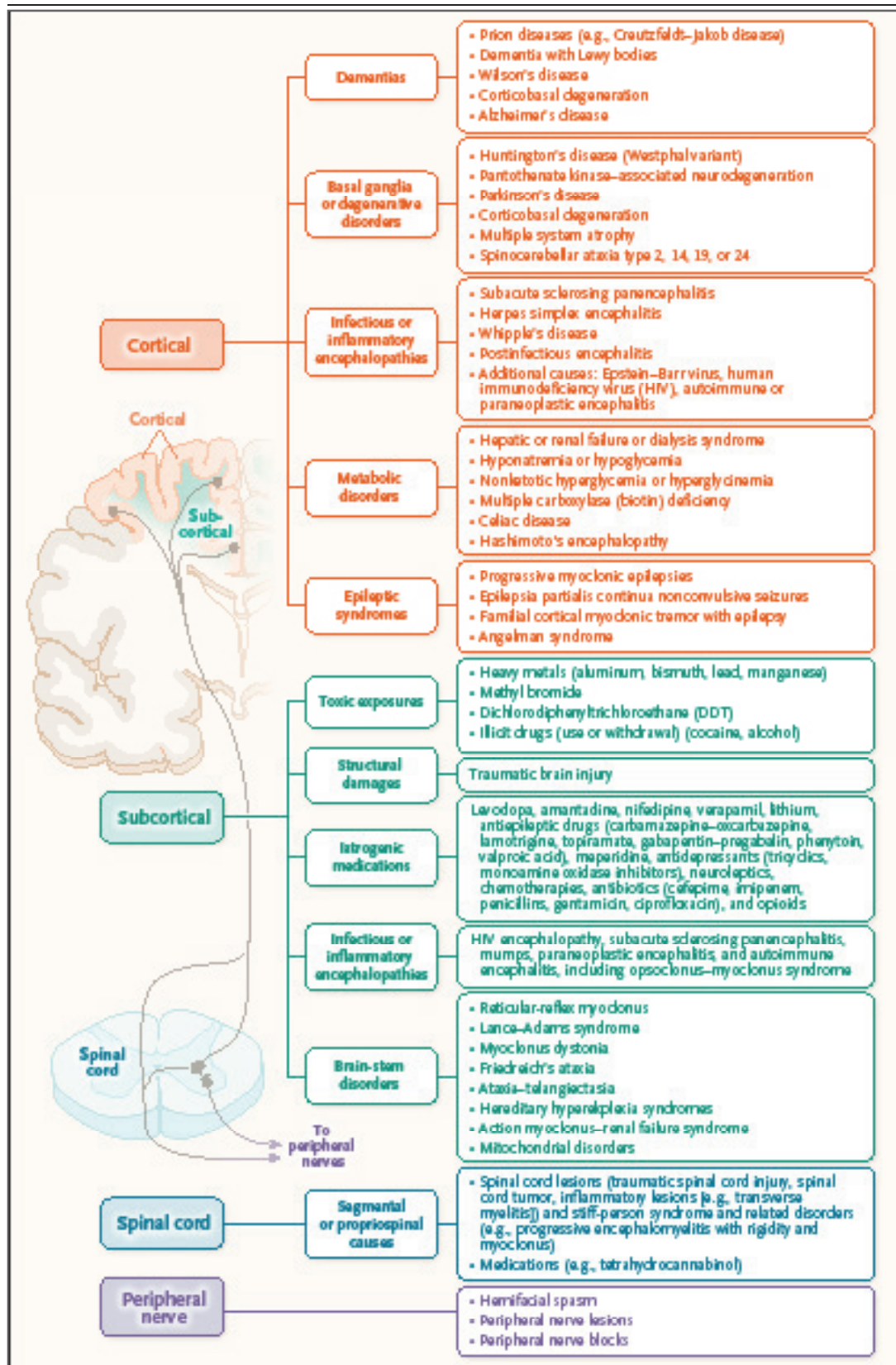


Figure 3 (facing page). Causes of Myoclonus.

This figure is adapted from Espay and Chen.⁹

than that seen in this case.¹⁹ The rapidity of this patient's disease progression, combined with the known diagnosis of Creutzfeldt–Jakob disease in her father, are highly suggestive of a genetic form of Creutzfeldt–Jakob disease.

DR. CHRISTOPHER D. STEPHEN'S DIAGNOSIS

Genetic Creutzfeldt–Jakob disease.

DIAGNOSTIC TESTING

Dr. Derek H. Oakley: A lumbar puncture was performed, and cytologic analysis of the CSF showed no evidence of pleocytosis or malignant cells. The CSF level of total tau was 1066 pg per milliliter (reference range, 0 to 1149), and CSF testing for 14-3-3 protein was negative. However, a real-time quaking-induced conversion (RT-QuIC) test for the detection of prion protein was positive; according to the Centers for Disease Control and Prevention diagnostic criteria for Creutzfeldt–Jakob disease,²⁰ this result is consistent with probable prion disease in the context of a neuropsychiatric disorder. A diagnosis of definite prion disease requires neuropathological or biochemical evaluation of brain tissue and thus was not possible at this time. The reference laboratory that performed the RT-QuIC test (the National Prion Disease Pathology Surveillance Center; NPDPS) interpreted the result as probable prion disease with a likelihood of greater than 98%.

ADDITIONAL FAMILY HISTORY

Dr. Stephen: Given the suspicion for Creutzfeldt–Jakob disease, additional history was obtained. The patient's sister reported that the patient's uncle (her father's fraternal twin) had also died from Creutzfeldt–Jakob disease, which provided further support for a diagnosis of hereditary Creutzfeldt–Jakob disease. There are striking similarities between the initial presentations of the patient and her relatives.

At 59 years of age, the patient's uncle began to have sensory symptoms in the left hand that were initially diagnosed as carpal tunnel syn-

drome. He subsequently had rapidly progressive weakness in his left hand, weakness in his right leg, and dysarthria. Spasticity with hyperreflexia developed, with loss of proprioception and vibratory perception, generalized ataxia, and cervical dystonia, and he was admitted to this hospital. Shortly thereafter, he began to have widespread myoclonic jerks and decreased responsiveness. Electroencephalography revealed general slowing and disorganization, with infrequent spike complexes. He died 4 months after the onset of symptoms in his left hand. An autopsy confirmed the diagnosis of Creutzfeldt–Jakob disease.

The patient's father began to have weakness in his right hand when he was 71 years of age, shortly after diagnosis of primary lung cancer. The weakness progressed to the right leg and the face, and dysarthria and dysphagia developed. Spasticity and hyperreflexia subsequently developed, with loss of temperature and vibratory perception in the feet and legs and ataxia on the right side. Electromyography suggested a motor axonal neuropathy. Akinetic mutism rapidly ensued after he was admitted to an affiliated hospital. Electroencephalography revealed the presence of bilateral spikes and polyspikes at irregular intervals. CSF evaluation was negative for 14-3-3 protein. MRI of the head revealed hyperintensities in the cortical ribbon on diffusion-weighted images involving the left paramedian cortex adjacent to the falx, insula, right anterior temporal lobe, and frontal lobes. He died 7 months after the onset of weakness. An autopsy confirmed the diagnosis of Creutzfeldt–Jakob disease.

Because the patient's father and uncle both had rapidly progressive weakness that began in an arm, followed by ataxia and myoclonus, and ultimately received a diagnosis of Creutzfeldt–Jakob disease that was confirmed by autopsy, blood samples were obtained from this patient and were sent to the NPDPS for genetic testing for pathogenic mutations of the prion protein gene (*PRNP*). Cases of monoparesis associated with genetic Creutzfeldt–Jakob disease are exceedingly rare; in a large case series reported by Cohen et al.,²¹ such a case was seen in only 1 of 77 patients.

GENETIC TESTING

Dr. Brian S. Appleby: Genetic prion disease is caused by point mutations or insertions or deletions in

an octapeptide region of PRNP. Genetic testing was performed in this patient as part of the comprehensive neuropathological evaluation conducted by the NPDPS (https://case.edu/medicine/pathology/divisions/prion-center/). Genetic analysis of PRNP was performed by the Center for Human Genetics Laboratory at University Hospitals Cleveland Medical Center; the protocol for this analysis has been described previously.²² The patient was found to have the E200K mutation and methionine homozygosity at codon 129, resulting in the E200K-129M haplotype. The patient's uncle had also undergone genetic analysis through the NPDPS and had been found to have the same haplotype (E200K-129M) and valine at codon 129 of the nonmutated allele. Genetic analysis was not available for the patient's father. E200K is the most common mutation that causes genetic Creutzfeldt-Jakob disease worldwide, with an age-related penetrance that approaches 100%.²³ Although most cases of prion disease with an E200K mutation are associated with a Creutzfeldt-Jakob disease phenotype, clinical heterogeneity has been noted.²¹

FOLLOW-UP

Dr. Stephen: The patient was discharged to a rehabilitation facility, where weakness in her right arm progressed to flaccid paralysis of the arm. The weakness in her right leg worsened, as did her dysarthria, and frequent myoclonus developed in both legs. She also described worsening vision, which greatly impaired her ability to see during communication through sign language. She became bedbound and required assistance to sit. Her genetic testing results were not yet available; however, those of her uncle were discussed with the family. Given that her presentation was similar to that of her father and uncle, there was a high suspicion for genetic Creutzfeldt-Jakob disease. After discussion with the patient and her family, the goals of care were changed to comfort measures. She died 5 months after the onset of weakness in her right arm, and the family agreed to an autopsy of the brain only. The patient's genetic testing results, which became available after her death, confirmed genetic Creutzfeldt-Jakob disease.

Figure 4 (facing page). Specimens of the Brain Obtained at Autopsy.

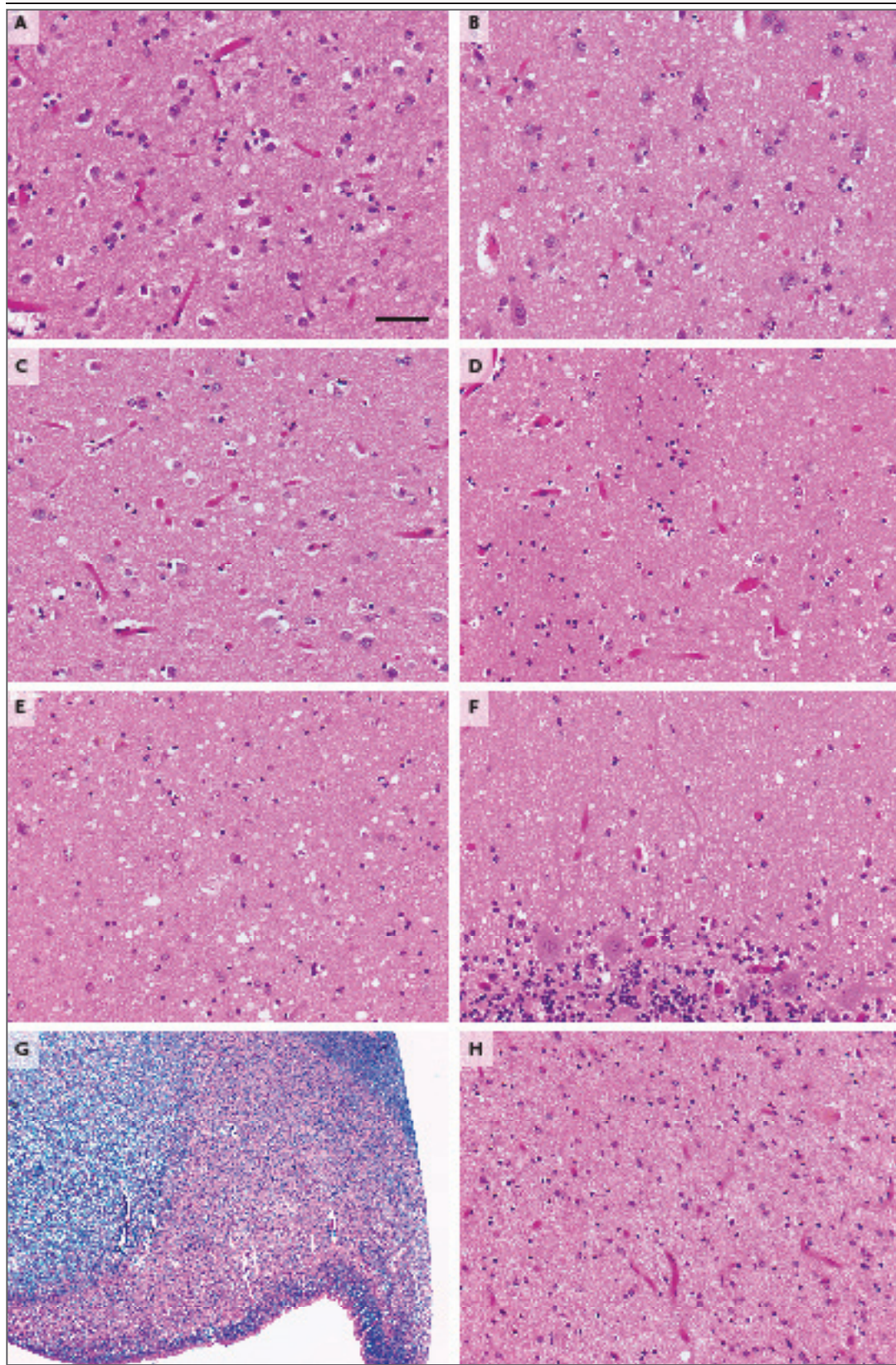
Hematoxylin and eosin staining (Panels A through F and Panel H) and Luxol fast blue–hematoxylin and eosin staining (Panel G) of the autopsy specimens were performed. Moderate transcortical spongiosis is present in the sampled areas of the cerebral cortex, including the middle frontal gyrus (Panel A), cingulate gyrus (Panel B), and perirhinal region (Panel C). Spongiosis is also present in the basal ganglia (Panel D) and thalamus (Panel E) and in the molecular cell layer of the cerebellum (Panel F). The lateral geniculate nucleus (Panels G and H) is depleted of neurons and shows severe gliosis (Panel H). The scale bar represents 50 μ m (Panels A through F and Panel H) and 500 μ m (Panel G).

AUTOPSY

Dr. Oakley: An autopsy of the brain was performed, with special precautions taken for the handling of prion disease tissue. Gross examination of the brain revealed mild atrophy in the frontal lobe but otherwise unremarkable cut sections. On microscopic examination, the diagnostic features of Creutzfeldt-Jakob disease were present (Fig. 4). There was mild-to-moderate spongiosis, neuronal loss, and gliosis in the sampled areas of the cerebral cortex. The deep gray nuclei showed patchy moderate spongiotic change and gliosis, as did the cerebellar cortex.

Gliosis and neuronal loss without spongiosis were noted in the lateral geniculate nucleus (Fig. 4). This finding is consistent with the patient's history of measles encephalitis and subsequent blindness rather than damage caused by prion disease. The primary visual cortex showed spongiosis consistent with prion disease but no evidence of chronic changes due to encephalitis. In addition, there was moderate arteriolar sclerosis, as well as minimal tau deposition that was restricted to the medial temporal cortex and pons.

Frozen and fixed brain tissue was sent to the NPDPS. Immunohistochemical analysis, which was performed with the use of an antibody to prion protein, revealed granular deposits, thus confirming the diagnosis of prion disease. In addition, Western blot analysis of the frozen brain tissue revealed the presence of abnormal protease-resistant prion protein, commonly referred to as PrP 27-30.



GENETIC COUNSELING AND FUTURE DIRECTIONS

Dr. Appleby: Given the hereditary basis of this disease, surviving family members are naturally anxious and concerned about the possibility of having Creutzfeldt–Jakob disease. Genetic counseling includes several key elements. Genetic testing for PRNP mutations is performed primarily on brain tissue at the time of autopsy or on blood. Persons wishing to know their genetic status can undergo a comprehensive assessment that includes genetic counseling as well as neurologic and psychiatric evaluations. The testing protocol for Huntington’s disease is strongly recommended for asymptomatic persons undergoing predictive genetic testing.²⁴ The disease and its lack of treatment options, as well as its autosomal dominant inheritance pattern and the penetrance of the specific mutation, if known, should be discussed.²³ Persons can choose to know the results or to have them withheld, including disclosure at a later date. Family planning is a common reason for the desire to know one’s mutation status. Reproductive options, including preimplantation genetic diagnosis with in vitro fertilization, should be discussed.²⁵ Fi-

nally, review of available patient support services and resources, such as those offered through the Creutzfeldt–Jakob Disease Foundation (<https://cjd.foundation.org/>), is recommended.

An increasingly common reason for genetic testing is the potential opportunity to participate in research and future clinical trials. Genetic prion disease is dependent on the production of prion proteins, and suppression of prion protein expression has the potential to affect the disease course, including the possibility of delaying or preventing the onset of disease in carriers of the mutation. An example of such an approach is the use of antisense oligonucleotides.²⁶ As experimental use of antisense oligonucleotides and genetic therapies become more common, it is likely that more persons will want to know their genetic status.

FINAL DIAGNOSIS

Genetic Creutzfeldt–Jakob disease.

This case was presented at Neurology Grand Rounds.

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Disclosure forms provided by the authors are available with the full text of this article at NEJM.org.

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